

CME396 – PROCESS PLANNING AND COST ESTIMATION

Unit 3 – Introduction to Cost Estimation

Important Questions with Answers (based on AM 2024, ND 2024, AM 2025, ND 2025 papers)

PART A – 2 MARK QUESTIONS WITH ANSWERS

1. Why is cost estimation important? What is its purpose? (AM 2024, AM 2025)

Cost estimation is the process of finding the probable cost of a product **before** it is manufactured. Its purposes are:

- To fix the selling price and submit quotations / tenders.
- To take make-or-buy decisions and select the most economical process.
- To prepare budgets and control the cost of production.

2. Compare estimation and costing. (ND 2025)

Estimation	Costing
Probable cost calculated before production.	Actual cost calculated during / after production.
Done by the estimator (technical person).	Done by the cost accountant.
Based on drawings, past data and judgement.	Based on actual bills, wages and records.
Used to fix price and submit quotations.	Used to control cost and find actual profit.

3. List the elements of cost. (AM 2025)

- **Material** – direct material and indirect material.
- **Labour** – direct labour and indirect labour.
- **Expenses** – direct expenses and indirect expenses (overheads).

Overheads are classified as factory, administrative, and selling & distribution overheads.

4. What are administrative expenses? (ND 2024)

Administrative expenses are the costs of running the office (not the shop floor). Examples: salaries of manager and office staff, office rent, stationery, telephone, postage, legal and audit fees. They are added to factory cost to get the production (office) cost.

5. List the methods of allocation of overhead expenses. (AM 2024, ND 2025)

1. Percentage of direct material cost
2. Percentage of direct labour cost
3. Percentage of prime cost
4. Man-hour rate method
5. Machine-hour rate method
6. Combination of man-hour and machine-hour rate
7. Unit rate method

6. A manufacturing concern produces 600 machines per year. If the total overheads during that year were Rs. 1,80,000, calculate the overhead cost per machine by unit rate method. (ND 2024)

$$\text{Overhead per machine} = \frac{\text{Total overheads}}{\text{Number of units produced}} = \frac{1,80,000}{600} = \text{Rs. 300 per machine}$$

7. Define man-hour rate and machine-hour rate. (ND 2024)

$$\text{Man-hour rate} = \frac{\text{Total factory overheads}}{\text{Total man-hours worked}} \quad \text{Machine-hour rate} = \frac{\text{Total cost of running the machine}}{\text{Total machine hours}}$$

Man-hour rate is used where work is mainly manual; machine-hour rate is used where work is mainly done on machines.

8. Define depreciation. List its methods.

Depreciation is the permanent decrease in the value of an asset with time due to wear and tear, age or obsolescence. Methods: (i) Straight line method, (ii) Reducing balance method, (iii) Sinking fund method, (iv) Sum-of-years' digits method, (v) Machine-hour basis method.

9. Define prime cost and factory cost.

- **Prime cost** = Direct material + Direct labour + Direct expenses
- **Factory cost** = Prime cost + Factory overheads

10. Differentiate direct and indirect material.

Direct material	Indirect material
Becomes a part of the finished product.	Helps production but does not become part of the product.
Can be traced to a particular job.	Cannot be traced to a particular job.
Example: steel bar for a shaft, castings.	Example: lubricants, cotton waste, grinding wheels.

PART B – 13 MARK QUESTIONS WITH ANSWERS

1. Explain the step by step procedure for doing cost estimation for a new product. (ND 2024 – 13a)

Definition

Estimation is the technical process of determining the probable cost of a product before its production starts, using its drawings and specifications.

Aims of estimation

- To fix the selling price and submit quotations or tenders.
- To decide whether to make or buy a component.
- To choose the most economical method and machine.
- To prepare production budgets and check the profitability of the product.

Procedure

1. **Study the drawing and specifications** – note material, dimensions, tolerances, surface finish and quantity.
2. **Prepare a bill of materials** – list every part, bought-out items and the quantity of each.
3. **Estimate material cost** – volume from the drawing $\rightarrow$ weight = volume $\times$ density $\rightarrow$ add allowances (machining, scrap, wastage) $\rightarrow$ weight $\times$ rate per kg.
4. **Decide the sequence of operations** – turning, drilling, milling, welding, painting, etc., and the machine for each.
5. **Estimate the time for each operation** – set-up time + machining time + handling time + allowances (fatigue, personal needs, tool change).
6. **Estimate labour cost** – time $\times$ wage rate per hour for each operation.
7. **Add direct expenses** – special tools, jigs, patterns, design or testing charges, royalties.
8. **Find prime cost** = material + labour + direct expenses.
9. **Add overheads** – factory overhead gives factory cost; administrative overhead gives production cost; selling & distribution overhead gives total cost.
10. **Add profit** – total cost + profit = selling price.

Example – cast iron pulley

Item	Working	Rs.
Material	$5 \text{ kg} \times \text{Rs. } 80/\text{kg}$	400
Labour	$2 \text{ h} \times \text{Rs. } 150/\text{h}$	300
Direct expenses	Share of pattern cost	50
Prime cost	$400 + 300 + 50$	750
Factory overhead	40% of labour	120
Administrative + selling overhead	$60 + 70$	130
Total cost	$750 + 120 + 130$	1,000
Selling price	$1,000 + 15\% \text{ profit}$	1,150

Conclusion

An accurate estimate helps the firm quote competitive prices, avoid losses on orders and control production cost.

2. Explain the elements of cost and the ladder of cost. Calculate prime cost, factory cost, production cost, total cost and selling price per item from the given data. (AM 2025, ND 2025 – 13a)

Elements of cost

1. Material

- Direct material – becomes part of the product (steel bar, castings, bought-out parts).
- Indirect material – helps production but cannot be traced to a job (oil, cotton waste, small tools).

2. Labour

- Direct labour – workers who directly work on the product (turner, welder, fitter).
- Indirect labour – supervisors, storekeepers, inspectors, maintenance staff.

3. Expenses

- Direct expenses – chargeable to one job (special jigs, patterns, hire of special machine, royalty).
- Indirect expenses (overheads) – rent, power, lighting, depreciation, insurance.

Overheads = Indirect material + Indirect labour + Indirect expenses.

Ladder of cost

1. Prime cost = Direct material + Direct labour + Direct expenses
2. Factory (works) cost = Prime cost + Factory overheads
3. Production (office) cost = Factory cost + Administrative overheads
4. Total cost = Production cost + Selling & distribution overheads
5. Selling price = Total cost + Profit

Prime cost	Direct material	Direct labour	Direct expenses
Factory cost	Prime cost		Factory overhead
Production cost	Factory cost		Admin. overhead
Total cost	Production cost		Selling & distribution
Selling price	Total cost		Profit

Fig. Ladder of cost (shaded = cost so far; open box = cost added; Selling price = Total cost + Profit)

Problem (ND 2025)

Cost of raw material in stock on 01-04-2023 = Rs. 25,000; raw material purchased = Rs. 40,000; direct labour = Rs. 14,000; direct expenses = Rs. 1,000; factory overhead = Rs. 9,750;

administrative expenditure = Rs. 6,500; selling and distribution = Rs. 3,250; number of items produced = 650; raw material in stock on 31-03-2024 = Rs. 15,000; net profit = 10% of total cost.

Solution

Step	Working	Total (Rs.)	Per item (Rs.)
Material consumed	$25,000 + 40,000 - 15,000$	50,000	76.92
Prime cost	$50,000 + 14,000 + 1,000$	65,000	100.00
Factory cost	$65,000 + 9,750$	74,750	115.00
Production cost	$74,750 + 6,500$	81,250	125.00
Total cost	$81,250 + 3,250$	84,500	130.00
Profit (10%)	0.10×130	8,450	13.00
Selling price	$130 + 13$	92,950	143.00

Answer: Selling price = Rs. 143 per item.

3. Explain the steps involved in estimating labour and material costs in manufacturing. (AM 2025 – 13a)

A. Estimation of material cost

1. Study the drawing and divide the component into simple shapes (cylinders, prisms, cones).
2. Calculate the volume of each shape and add them (subtract the volume of holes).
3. Weight = Volume $\times$ Density (steel 7.86 g/cm³, cast iron 7.2 g/cm³, aluminium 2.7 g/cm³).
4. Add allowances – machining allowance, sprue / scrap loss, cutting (kerf) loss, scale loss in forging.
5. Material cost = Gross weight $\times$ Rate per kg + cost of bought-out parts.
6. Subtract the scrap value if the scrap can be sold.

$$\text{Material cost} = (\text{Volume} \times \text{Density} + \text{Allowances}) \times \text{Rate per kg} - \text{Scrap value}$$

B. Estimation of labour cost

1. List the operations from the process (operation) sheet.
2. Find the time for each operation:
 - **Set-up time** – time to prepare the machine, fix jigs and tools.
 - **Operation (machining) time** – from formulas or time study.
 - **Handling time** – loading, unloading and measuring the job.
3. Add allowances of about 10–20% for fatigue, personal needs, tool sharpening and delays.
4. Find the total time per piece:

$$\text{Time per piece} = \frac{\text{Set-up time}}{\text{Batch size}} + \text{Operation time} + \text{Handling time} + \text{Allowances}$$

$$\text{Labour cost} = \text{Total time} \times \text{Wage rate per hour}$$

Factors affecting labour cost

Skill of the worker, type of machine, batch size, wage rate, incentive schemes and working conditions.

Example

A shaft needs 12 min turning + 3 min handling, with 15% allowance. Total time = $15 \times 1.15 = 17.25$ min. At Rs. 120 per hour:

$$\text{Labour cost} = \frac{17.25}{60} \times 120 = \text{Rs. 34.50 per piece}$$

Conclusion

Material and labour together form the prime cost, so accurate estimation of both is the base for a correct selling price.

4. Describe overheads and the various methods of allocating overhead charges in cost estimation. (AM 2024, AM 2025 – 13b, ND 2025)

Definition

Overheads are all indirect costs – indirect material, indirect labour and indirect expenses – which cannot be charged directly to a particular job. They are distributed to all jobs on a fair basis.

Classification of overheads

Type	Examples
Factory (works) overhead	Factory rent, power, lighting, repairs, depreciation of machines, supervisor salary
Administrative (office) overhead	Office staff salaries, office rent, stationery, telephone, audit fees
Selling overhead	Advertisement, salesmen's salary and commission, showroom expenses
Distribution overhead	Packing, warehouse rent, transport and delivery vans

Methods of allocation

1. Percentage of direct material cost – used when material is the main cost.

$$\% \text{ Overhead} = \frac{\text{Total overheads}}{\text{Total direct material cost}} \times 100$$

2. Percentage of direct labour cost – simple; used when wage rates are uniform.

$$\% \text{ Overhead} = \frac{\text{Total overheads}}{\text{Total direct labour cost}} \times 100$$

3. Percentage of prime cost – considers both material and labour.

$$\% \text{ Overhead} = \frac{\text{Total overheads}}{\text{Total prime cost}} \times 100$$

4. Man-hour rate – time based; suitable for manual work.

$$\text{Man-hour rate} = \frac{\text{Total overheads}}{\text{Total man-hours}}$$

5. Machine-hour rate – the best method for machine shops.

$$\text{Machine-hour rate} = \frac{\text{Total machine running cost}}{\text{Total machine hours}}$$

6. Combination of man-hour and machine-hour rates – used where both men and machines are important.

7. Unit rate method – used when all products are identical.

$$\text{Overhead per unit} = \frac{\text{Total overheads}}{\text{Number of units produced}}$$

Example

Total overhead = Rs. 50,000 and total direct labour = Rs. 1,00,000, so overhead = 50% of labour. A job with Rs. 2,000 labour is charged Rs. 1,000 overhead.

Conclusion

Machine-hour rate is the most accurate method in modern machine shops, because most overheads are caused by machines.

5. Explain depreciation, its causes and methods, with a note on appreciation. Solve the given problems. (ND 2024 – 13b, ND 2025 – 13b)

Definition

Depreciation is the permanent decrease in the value of an asset (machine, building, tool) with time. It is charged as a cost so that money is available to replace the asset at the end of its life.

Causes

- Wear and tear due to use.
- Age and decay (rust, corrosion).
- Obsolescence – a better machine becomes available.
- Accidents and breakdowns; fall in demand for the product.

Methods (C = initial cost, S = scrap value, n = life in years)

1. Straight line method – equal amount every year.

$$D = \frac{C - S}{n}$$

2. Reducing balance method – fixed percentage of the remaining value each year.

$$r = 1 - \left(\frac{S}{C}\right)^{1/n} \quad \text{Value after } m \text{ years} = C(1 - r)^m$$

3. Sinking fund method – a fixed sum A is invested every year at interest rate i.

$$A = (C - S) \times \frac{i}{(1 + i)^n - 1}$$

4. Sum-of-years' digits method

$$D_k = (C - S) \times \frac{n - k + 1}{n(n + 1)/2}$$

5. Machine-hour basis – fair when machine use varies.

$$D \text{ per hour} = \frac{C - S}{\text{Total working hours in life}}$$

Appreciation

Appreciation is the increase in the value of an asset with time. Land, buildings in growing cities and rare materials usually appreciate. Example: a factory plot bought for Rs. 10 lakh may be worth Rs. 15 lakh after 5 years.

Problem (i) – Reducing balance method (7 marks)

A machine costs Rs. 25,000, life 20 years, scrap value Rs. 4,000. Find the depreciation fund after 3 years, depreciation for the 3rd year and the percentage by which the value is reduced every year.

$$r = 1 - \left(\frac{4,000}{25,000}\right)^{1/20} = 1 - (0.16)^{0.05} = 1 - 0.9124 = 0.0876 = \mathbf{8.76\%}$$

[Using logs: $\log 0.16 = -0.7959$; $\times 0.05 = -0.0398$; $\text{antilog} = 0.9124$]

Year	Value at start (Rs.)	Depreciation @ 8.76% (Rs.)	Value at end (Rs.)
1	25,000.00	2,188.90	22,811.10
2	22,811.10	1,997.24	20,813.86
3	20,813.86	1,822.33	18,991.53

Depreciation fund after 3 years = 25,000 – 18,991.53 = Rs. 6,008.47

Depreciation for 3rd year = Rs. 1,822.33 Rate of reduction = 8.76% per year

Problem (ii) – Machine-hour basis (6 marks)

A shaper costs Rs. 85,000, life 8 years, scrap value Rs. 8,500, works 3,330 hours a year.

$$\text{Total hours} = 8 \times 3,330 = 26,640 \text{ h}$$

$$\text{Depreciation per hour} = \frac{85,000 - 8,500}{26,640} = \frac{76,500}{26,640} = \text{Rs. 2.87 per hour}$$

$$\text{Annual depreciation} = 2.8716 \times 3,330 = \text{Rs. 9,562.50 per year} (= 11.25\% \text{ of cost})$$

6. Discuss the various methods of costing such as job costing, process costing and activity-based costing, highlighting their differences and suitability. (AM 2024 – 13a)

Introduction

The method of costing depends on the type of product and the way it is manufactured.

Methods

1. **Job costing** – cost is found separately for each job or order. Used in job shops, repair shops and special machines made to order. Each job has its own cost sheet.
2. **Batch costing** – a batch of identical items is treated as one job.

$$\text{Cost per item} = \frac{\text{Total cost of batch}}{\text{Number of items in batch}}$$

Used for bolts, gears, medicines, toys.

3. **Process costing** – used where the product passes through several processes in sequence (chemicals, cement, sugar, steel, paper). Cost is collected for each process and transferred to the next.
4. **Unit (single output) costing** – one standard product in large quantity (bricks, cement bags, mining).
5. **Operating costing** – for services: transport (cost per km), power (cost per kWh), hospitals.
6. **Contract costing** – for large, long-duration works such as buildings, bridges and ships; each contract is a cost unit.
7. **Activity-based costing (ABC)** – overheads are traced to activities (set-ups, inspection, material handling) and charged to products based on how much of each activity they use. More accurate where overheads are high and products vary.

Comparison

Point	Job costing	Process costing	Activity-based costing
Type of product	Made to order, different each time	Continuous, identical	Many products with high overheads
Cost unit	One job	One process / period	One activity
Overhead charged	Broad rate	Per process	Per activity (cost driver)
Accuracy	Moderate	Average cost only	High
Example	Machine repair	Cement plant	Auto parts plant with many models

Conclusion

Job costing suits job-order shops, process costing suits continuous production, and ABC gives the most accurate product costs in modern multi-product factories.